# CSA0609-DESIGN & ANALYSIS OF ALGORITHM
# CO1 - AT1: ALGORITHM SCALABILITY AUDIT
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## 1. Problem Statement & System Context

ShopSphere is a rapidly growing online shopping platform transitioning its inventory engine from a catalog of **N = 100,000** items to a enterprise-scale catalog of **N = 1,000,000** items. The platform's backend infrastructure executes operations on a dedicated processing node with a sustained execution rate of **10^8 basic operations/second (100 MHz effective rate)**. All batch algorithms must complete within a strict nightly SLA processing window of **10.0 minutes (600 seconds)**.

## 2. Computational Modules Overview

### Module M1: Duplicate Product Detector

Executes a pairwise comparison across the catalog to identify duplicate SKUs.

Pseudocode:

```
Algorithm DuplicateDetector(P[0..n-1]):
   for i = 0 to n - 2 do:
      for j = i + 1 to n - 1 do:
         if P[i].SKU == P[j].SKU then:
            FlagDuplicate(P[i], P[j])
```

### Module M2: API Rate Limiter Audit

A recursive audit function that logs API key activity across rate-limiting buckets.

Pseudocode:

```
Algorithm AuditRateLimiter(n):
   if n <= 1 then:
      LogAccess()
      return
   LogAccess()
   AuditRateLimiter(floor(n / 2))
   AuditRateLimiter(floor(n / 2))
```

### Search Subsystem (S1 & S2)

S1: Sequential Search — scans unsorted user cart items sequentially. Success probability p = 0.5.
S2: Binary Search — searches the sorted global catalog array for matching item IDs.

### Module M4: Product Recommendation Engine

Current Algorithm: Divides catalog into 4 subproblems of size n/2, with linear merge work n.
Recurrence: $T(n) = 4\,T(n/2) + n$
Proposed Redesign: Divides catalog into 2 balanced subproblems of size n/2, with linear merge work n.
Recurrence: $T(n) = 2\,T(n/2) + n$

### Module M5: Analytics & Reporting Engine

Analytics Data Engine: Processes event streams via divide-and-conquer recurrence:
$T(n) = 2\,T(n/2) + n \log_2(n)$
Nightly Report Generator: Performs full matrix cross-referencing taking quadratic time:
$T(n) = 3\,n^2 + 10\,n \log_2(n) + 50$ ops

## 3. Assessment Questions (Tasks a – g)

**(a) Input & Basic Operation Identification:** For Module M1 and Module M2, identify the input parameter n, the basic operation executed, its exact location in code, and state whether basic operation count varies for inputs of the same size n.

**(b) Exact Mathematical Derivation:** Derive the exact basic operation count formulas C(n) for M1 and A(n) for M2. For M2, state the exact formula for $n = 2^k$ and general n. Evaluate A(1,024) and verify against formula A(n) = n-1. Critically evaluate a team member's claim that M2 is Theta(n log n).

**(c) Recurrence Analysis for Binary Search S2:** Set up the recurrence relation for Binary Search S2. Solve it using backward substitution to determine asymptotic growth. Compare auxiliary memory space for iterative O(1) vs recursive O(log n) implementations at n = 1,000,000.

**(d) Master Theorem & Recursion Tree Analysis:** Apply the Master Theorem to analyze Module M4 (Current vs Proposed). For Module M5 Analytics Engine $T(n) = 2\,T(n/2) + n \log_2(n)$, explain why Master Theorem cannot be applied directly and derive exact asymptotic bound using Recursion Tree method.

**(e) Best, Worst, and Average Case Analysis for S1:** Define best, worst, and average-case complexities for Sequential Search S1. Derive average comparisons $C_{avg}(n)$ as a function of target presence probability p. Calculate exact comparisons for n = 1,000,000 and p = 0.5.

**(f) Rigorous Asymptotic Proofs:** Using formal definition of Big-Theta, prove that $f(n) = 3\,n^2 + 10\,n \log_2(n) + 50$ is $\Theta(n^2)$. Find explicit positive constants c1, c2, n0. Prove using limits that $f(n) \neq O(n)$.

**(g) Empirical Scalability Audit & SLA Evaluation:** Construct a master operations and execution time table for all modules at N = 100,000 and N = 1,000,000 at server rate 10^8 ops/s. Identify all SLA failures (600s threshold) and provide architectural redesign recommendations.

# 4. Evaluation Rubric

| Criteria | Weight | Exemplary (Level 4) | Unsatisfactory (Level 1) |
|---|---|---|---|
| Mathematical Derivations | 25% | Flawless summation and recurrence derivations with exact constants. | Incorrect summations or missing steps. |
| Asymptotic Proofs | 25% | Rigorous c1, c2, n0 constants and formal limit test proofs. | Vague claims without formal definitions. |
| Scalability Audit | 25% | Exact operation counts and SLA failure diagnosis at 10^8 ops/s. | Inaccurate counts or incorrect SLA conclusions. |
| Architectural Redesign | 25% | Clear, viable redesign strategies (Hash Maps, streaming, 2-way split). | Superficial recommendations without complexity improvements. |